

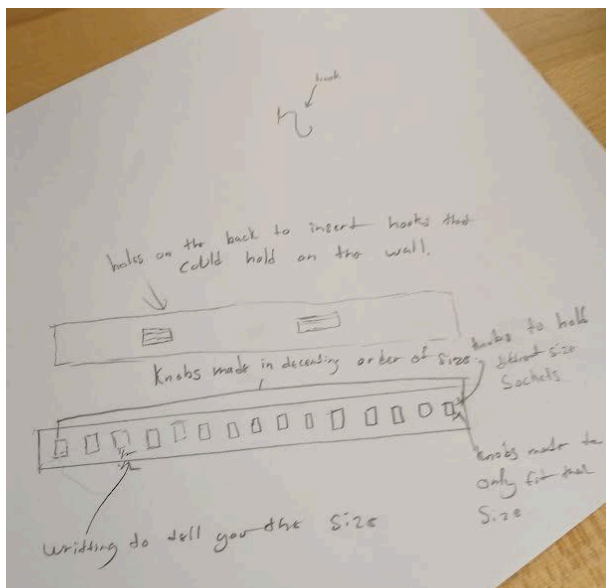
MyDesign Portfolio

Element D Initial Template

2 organizing sticks

Date: 3 /17/2025

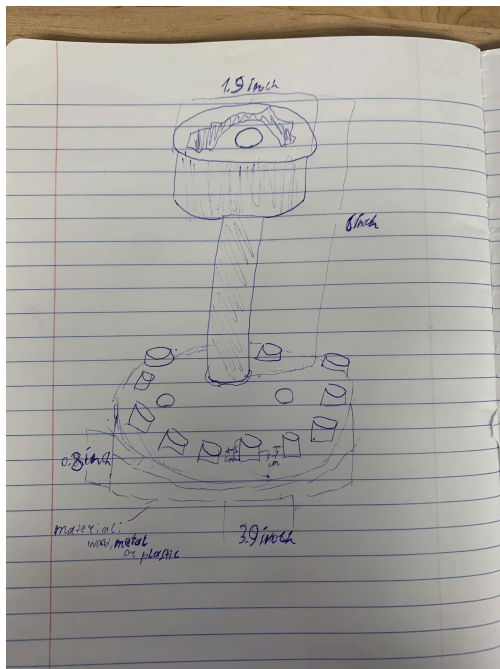
Description: A long white, 3D printed stick with holes that have centerpieces that have the ability to hold down sockets based on their size. Holes on the back are able to fit in 3D printed hooks right in the back to be able to hang on hole boards.



Vertical Rotating Socket Holder

Date:3/19/2025

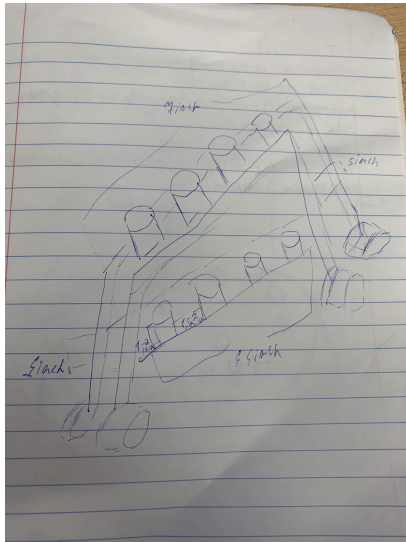
Description:A circular stand which can rotate with slots where sockets will be inserted with size labels to know where sockets need to be. It makes it easy to take the socket size which you want.



Magnetic Rail Holder

Date:3/19/2025

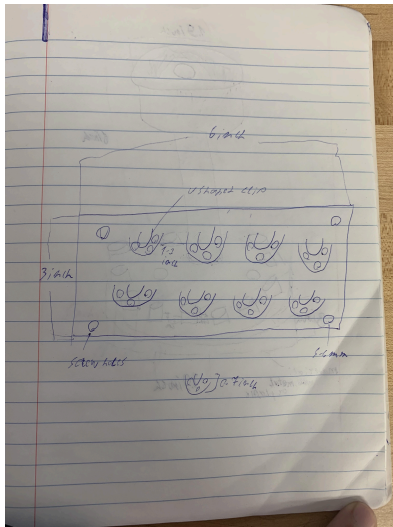
Description: A rail with magnetic base which will hold sockets firmly on each place and each socket can be fixed with an additional clamp or just held by magnetic. Also size labels will help to find what you need.



Clip on socket holder

Date:3/19/2025

Description: Socket holder which will be with a row of u-shaped clip that will hold sockets. The clips allow each socket to snap in securely. Also for example these clips can be attached to walls.

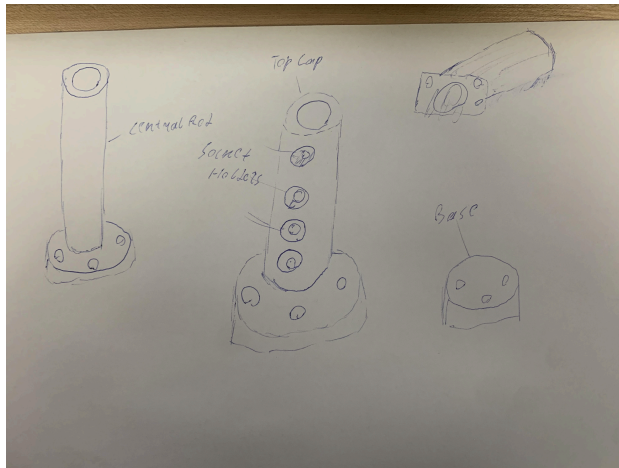


Vertical Tubular socket holder



Date:3/19/2025

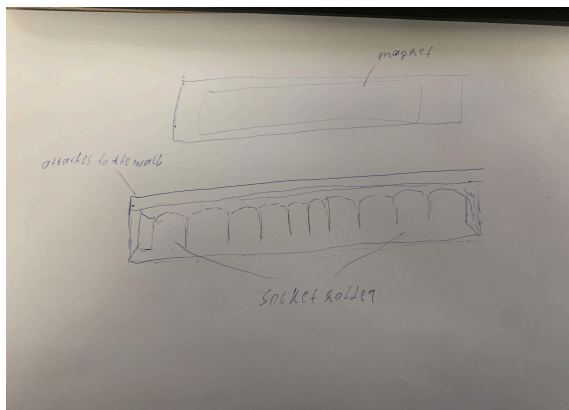
Description: A cylindrical tube with some spaced holes for holding socket heads, which mounted on a circular base for stability. The holder can be made of 3d printing. Also this tube will be compact and will save space on the table.



Socket Strap Holder

Date:3/19/2025

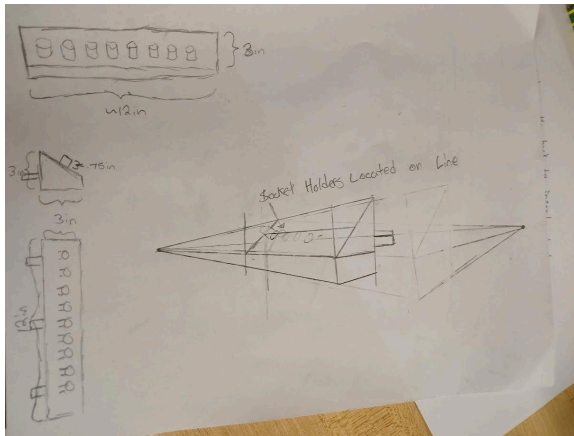
Description: A holder which will look like a strap and have a magnet on the back. That magnet will help attached to the walls which make this holder compact and easily to find what you want



Angled Horizontal Socket Holder

Date: 3/19/2025

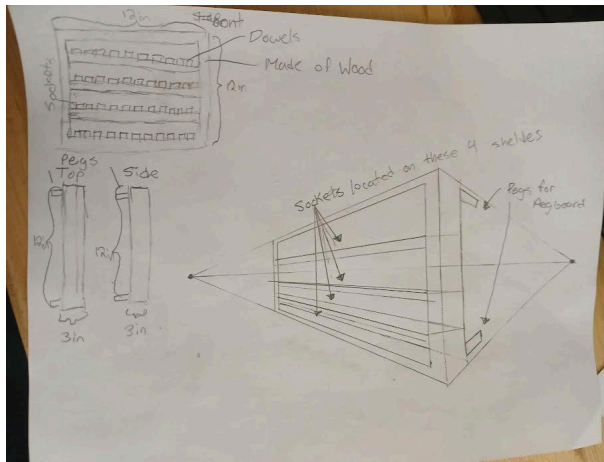
Description: Multiple long rectangular Holders that can be placed on a pegboard via 2 dowels on each end are also at a 45 degree angle facing the way people would interact with the board. These are modular so if more sockets are acquired we can print them again to match the new amount, there will also be space for labels below the socket holder. (The object is to be made of 3d printed PLA)



4 Level Shelf

Date: 3/21/2025

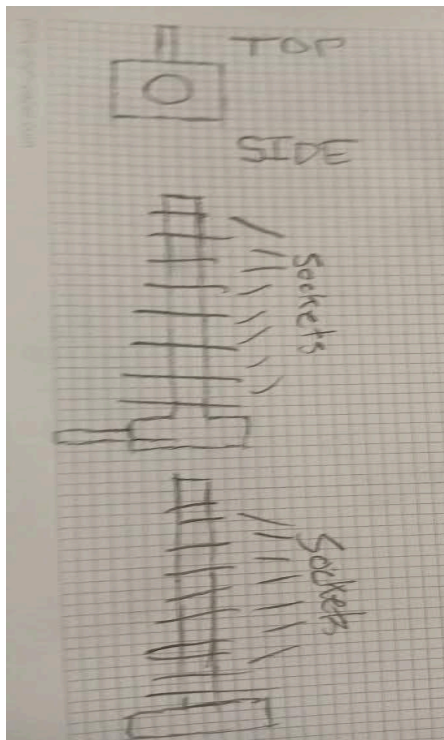
Description: A shelf like design in which each "floor" of the shelf can hold 12 sockets. The dimensions are roughly 12x12x3 in the design is made of wood and in each of the four corners facing away from the front dowels will be used to hold the design on the pegboard allowing for ease of removal and storage. The design would be painted a desired color afterward



Tower Socket Holder

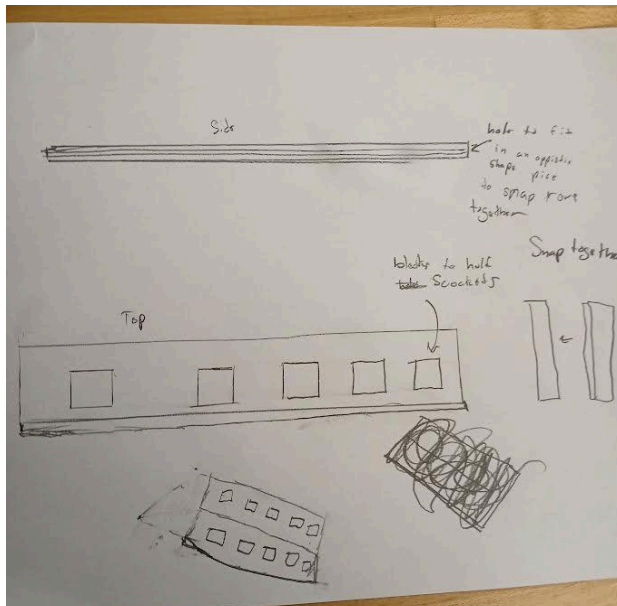
Date: 3/25/2025

Description: A single long dowel which holds the sockets in whatever direction is most convenient for the pegboard in which the object is being designed. This design will function by stacking the sockets



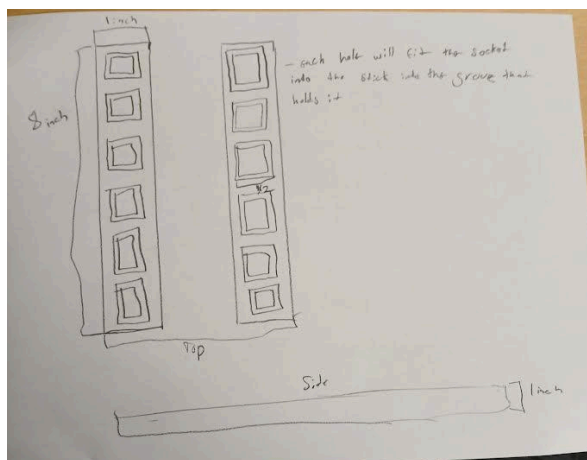
Date: 3/21/2025 *Connector Piece strips*

Description: A two piece holder that can hold 5 sockets on each block using blocks that fit inside the sockets in respect to the sockets specific size. The two rows can snap together to form one big block holding sockets in a 2x5 pattern.



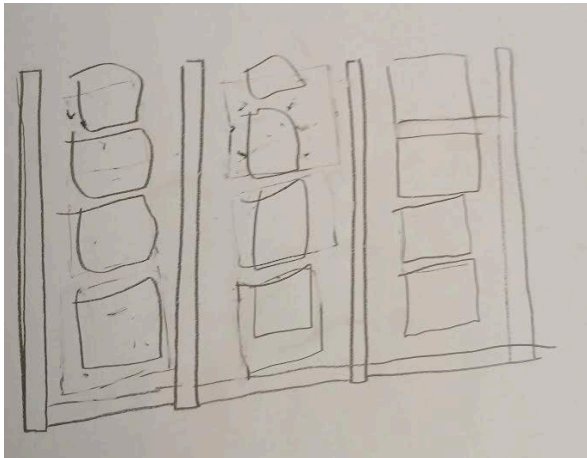
Date: 3/21/2025 *Socket Molds*

Description: Two rectangles with an 8 inch length that can hold six sockets on each. The sockets are secured by indents in the rectangles that are able to fit the socket's shape in with around 1/16 inch error of margin



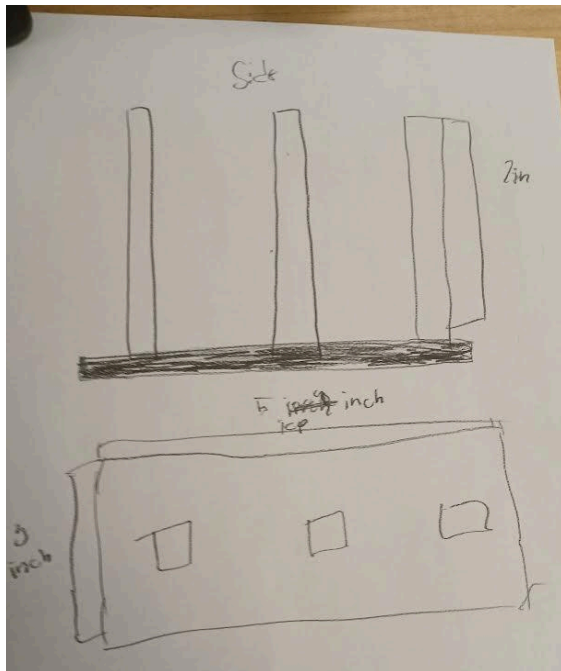
Date: 3/21/2025 *Connect Four*

Description: A set of tall rectangular prisms that fit the sockets vertically in stacks of four sockets. Each prism is connected to one another and there are 3 of them. Prisms will be made to fit each socket size.



Date: 3/21/2025 *Tower Holders*

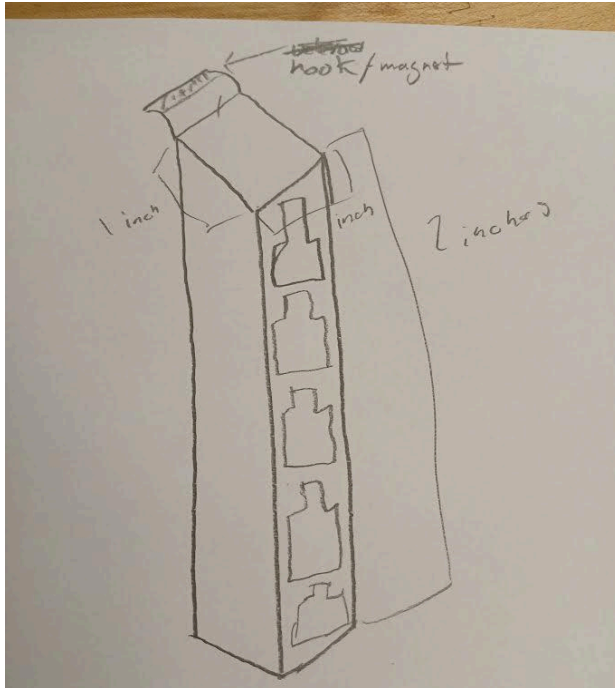
Description: A set of towers that are able to fit in the center of the sockets. There are three of the towers and they can stack 5 sockets vertically. They have a base a high of 1 inch and a width of 3 inches and a length of 5 inches.



Date: 3/21/2025 *Hook on box*

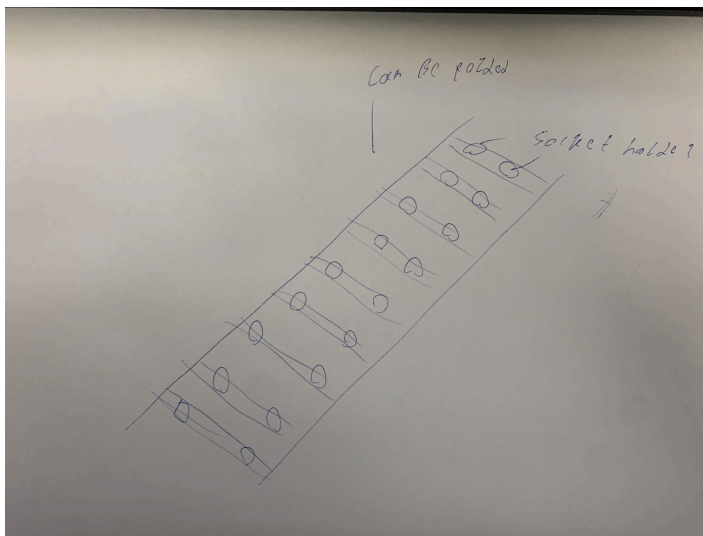
Description: A rectangular box that can hook on to your hip and can hold five sockets. The sockets go into the box vertically. Can make as many as we want. Will be made out of wood

or 3D material. The hook will be made with a magnet built into it.



Date: 3/21/2025 Socket Rails

Description: socket holder organizer in the form of rails where each strip has 2 socket holders. Also this design can be folded or rolled up which makes it convenient.



Design Idea Comparison and Selection (D1, D2)

Design Ideas	Simplicity to Design (Weight = 2)	Cost (Weight = 2)	Weight / Heaviness (Weight = 2)	Ease of use (Weight = 3)	Size (Weight = 3)	Ease of Replication (Weight = 1)	Ease to Construct (Weight = 2)	Efficient Storage (Weight = 3)	Total
<i>4 Level Shelf like design which attaches to the pegboard with 4 peg in the 4 corners</i>	2/5	2/5	1/5	3/5	1/5	1/5	1/5	4/5	37/90
<i>Angled Horizontal Socket Holder which attaches directly to pegboard</i>	4/5	4/5	4/5	5/5	3/5	5/5	5/5	2/5	69/90



2 Organizing Sticks which have pegs on them which attach to the pegboard	5/5	3/5	4/5	4/5	3/5	4/5	4/5	3/5	66/90
Clip on Socket Holder: This holder attaches to the pegboard and has U-shaped clips to hold the pegs	3/5	4/5	2/5	3/5	2/5	1/5	2/5	3/5	47/90
Tower Socket Holder which is vertical on the pegboard	3/5	4/5	3/5	2/5	3/5	3/5	3/5	2/5	50/90

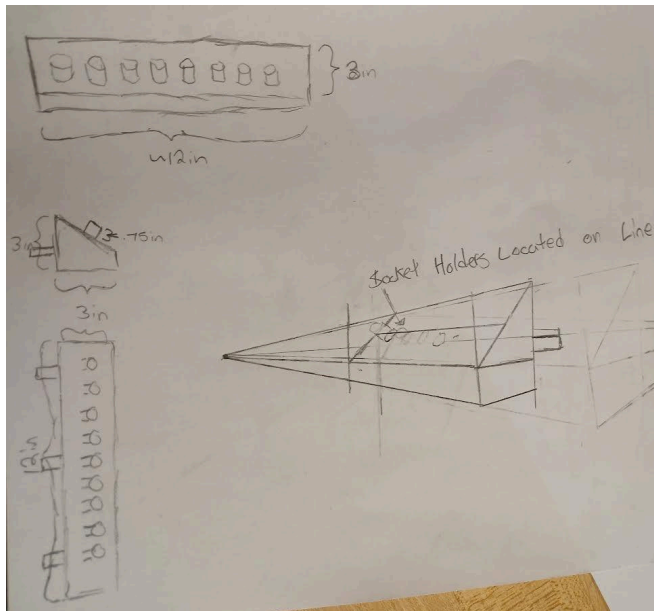
Describe all of your design's details such that someone unfamiliar with your project could build your prototype. Include sketches, dimensions, CAD (if taught in your class), manufacturing methods, material choices, supply lists, and costs.

Design Blueprint #1



Date: 3/27/2025

Sketch:



Predicted weight is 0.12 lbs

Manufacturing Methods:

Step 1: 3D print the base of the holder and sand the design for magnet fitting and aesthetics (not including pegs)

Step 2: install magnets into the back corresponding to the holes for them

Step 3: 3D print the pegs which are then fitted with magnets

Supply List and Costs:

Item	Cost each	# needed	Total
3D Printed filament	0	0.15lbs	0
Magnets	\$0.15	7-8	\$1.05-\$1.20
Total Cost			\$1.05-\$1.20

Plan of Action (D3)

A Gantt chart is encouraged to lay out the primary steps expected on a visual timeline. Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document. Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.

Socket holder					
	1	2	3	4	5
Sketch	X				
CAD	X	X			
Print and assemble		X			
Test fit sockets		X	X		
Test capacity			X		
Test durability			X		
Misc			X	X	
sanding				X	
Test new sizing				X	X
Finishing					X